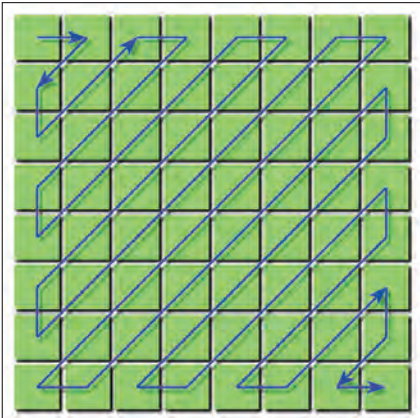


JPEG2000

Successor to JPEG, version 2000 is effective with higher image resolutions, but fails with lower resolutions

PNG

Is an image compression format developed in the early 90s to replace GIF (graphics interchange format)

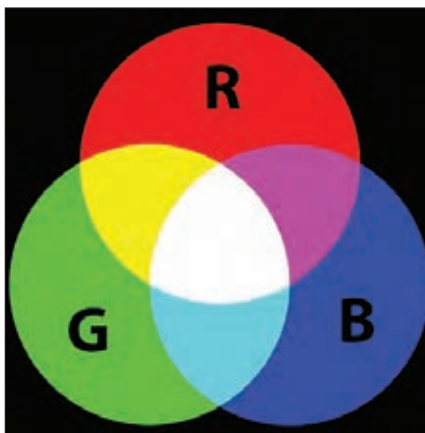


The zigzag order of JPG picture segments

picture, but for information such as date and time of shooting. The segments of the image are in a zig-zag order starting at the top left corner of the image.

When a bitmap image is opened in an application such as Microsoft Photo Editor, or Adobe Photoshop and saved as a JPEG, there is a reduction in file size by up to 90 per cent, with no perceivable drop in visual quality. Let us see how this is done.

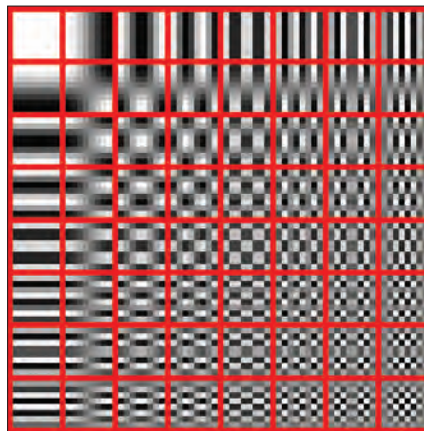
1. The bitmap image, with each block of 8 bits (or 16 or 32 bits, depending upon colour quality) standing for a different block, is converted to an RGB image, i.e. a method of representation in which each colour, instead of being represented by a number from 1 to 255, is represented by the red, blue and green contributions to it. The Red Green Blue Representation of colours for a pixel. A pixel colour is specified by specifying the percentage compo-



A pixel colour is specified by the percentage composition of the colour in terms of red, green and blue

sition of the colour in terms of Red, Green and Blue colours.

2. The RGB image is converted into a YCbCr image, i.e. an image in which, instead of specifying the three colour contributions of a pixel, its brightness (Y), and two components of colour (Cb and Cr) are specified.
3. The resolution of the Cb and Cr components are reduced, often almost to half its original value i.e. many nearby pixels are given the same average colour. This is because the human eye is more sensitive to changes in brightness than changes in colour.
4. The image is split into 8x8 pixel sectors, and statistical techniques such as discrete cosine transforms and discrete fourier



Each pixel is expressed as a linear combination of these 64 squares, to enable better compressions.

transforms are applied depending upon whether the colour gradient across them is gradual or sudden.

5. The image is re-specified in terms of similar pixels, their frequency of occurrence and their locations.
6. The resolution of the specifications for very low frequency pixels is reduced. In this way, the image is taken through multiple types of representations and unnecessary information is eliminated. Most JPEG codecs allow you to choose the compression ratio you desire, so that the trade-off between size and quality is your decision.

MP3

One of the most popular forms of lossy compression used in audio files world-

wide is MP3. Developed in the '80s by the Moving Picture Experts Group (consisting of engineers from Fraunhofer IIS, Germany and Alcatel-Lucent, New Jersey,) MP3 stands for MPEG 1, Audio Layer 3.

This method of compression was initially developed by MPEG as a method of compression for the audio in MPEG 1 video files, the format used in VCDs, but soon found independent application.

The MP3 file structure

An MP3 file consists of various frames, one after the other. Each frame consists of an MP3 header, and the actual data. In an MP3 file, the actual data in each frame is 1152 samples long and is known as a block.

A sample is a discrete digital signal that is generated by a microphone when audio is incident on it. The MP3 header contains the essential information required to play the block while the block contains the actual audio data.

A header and a block together make up a frame of the MP3 file and is of variable duration in playback, which is specified in the header. The header contains the following information.

1. A sync word, data to make sure the audio and video are synchronised (MP3 was originally meant to be used on VCDs)
 2. Version of the MP3 codec
 3. **Layer:** The number of the layer to which it belongs
 4. **Bit rate:** The bit rate (in kbps) used by that specific block.
 5. **Frequency:** The speed with which the samples in the block are to be played.
 6. **Error protection:** A reference to a similar audio block somewhere in the file that can be substituted for the current one, if the current one cannot be played due to errors. More technical information is specified in the header.
- When you open an MP3 file using a player, it reads the first frame, opens the header, plays the block associated with the header according to the specifications (in the header), moves to the next frame and then repeats the process.

Sometimes the first block in an MP3 file is an ID3 tag. ID3 is a universally accepted format by which miscella-